
8.1 OBJECTIVES

This chapter would make you understand the following concepts

- Why is Email Security needed?
- How does Pretty Good Privacy ensure Email Security
- S/MIME provides Security
- How to secure transmission of packets at network layer using IP protocol
- How to ensure confidentiality and authentication

8.2 INTRODUCTION

Electronic mail is commonly known as E-mail or Email is a technique of exchanging information(mail) between people using any electronic devices. In 1971, Ray Tomlinson invented networked email. He developed the first system which made it possible to send mail between people/organizations using different devices across the Internet, using the @ symbol which links the username with the destination device.

By the mid-1970s, it gained its popularity as email.

Nowadays, Emails are being used by everyone, maybe for professional or personal use which may even carry confidential information.

Hence, it becomes very essential to provide security to emails as well.

Email security ensures us that there is control over the content of any email account.

An email service provider enables efficient email security to protect any email accounts and its data from attackers or hackers and deliver the mail to a valid recipient.

8.3 PRETTY GOOD PRIVACY (PGP)

Prior to the Pretty Good Privacy technique was introduced, the email provider, Internet Provider, hackers, or even the government could view and read your messages.

But later, PGP was developed in the 1990s by Phil Zimmermann, allowing email and any other types of messages to be exchanged securely.

In today's time, PGP has already been standardized into OpenPGP, which enables anyone to write Pretty Good Privacy (PGP) software which is interoperable and compatible with other implementations.